

Cross-Silo Federated Learning

for Privacy-Preserving Medical Image Classification

Training a skin-lesion classifier across 6 hospitals — without ever sharing patient images

Mahmoud Asadi Heris · Bachelor Creative Technologies & AI (Howest) · Innovation & Research Project

The problem & the research question

Valuable medical data is locked inside institutions by law and ethics

- Hospitals hold exactly the data needed to train good diagnostic AI.
 - Privacy law + ethics mean patient images usually cannot leave the building.
 - So how do several hospitals train one strong model together — sharing no raw data?
-
- Benchmark: Fed-ISIC2019 (FLamby) — ~23,000 dermoscopy images, 8 diagnoses, 6 hospitals.
 - Metric: balanced accuracy (all 8 classes count equally; random = 0.125).

Research question

How can cross-silo Federated Learning be engineered to train clinically relevant image-classification models without sharing raw data — while quantifying the trade-offs between performance, data heterogeneity, and privacy?

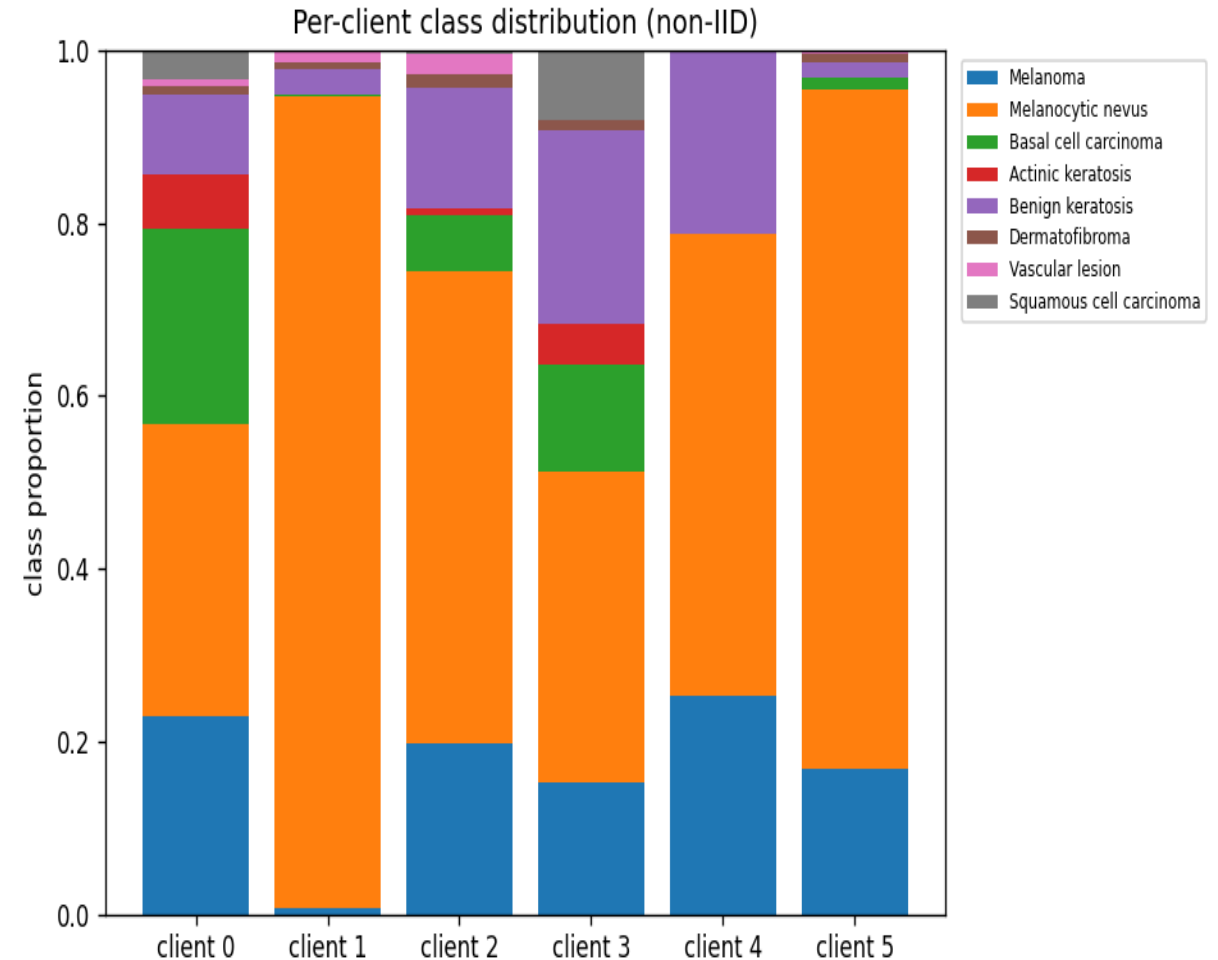
Federated Learning: each hospital trains locally and sends only model updates (numbers) to a coordinator that averages them. The images never move.

The data is highly uneven across hospitals (non-IID)

This unevenness is exactly what makes federated learning hard

- 6 hospitals, sizes from 7,947 down to 281 images.
- Some hospitals are missing whole disease classes.
- Each bar (right) is one hospital; colours are the 8 diagnoses — the mixes clearly differ.
- We quantified this: label entropy, missing-class counts, and Jensen–Shannon distance from the global mix.

If every hospital looked the same, averaging their models would be trivial. Because they don't, naive averaging can wobble — motivating every method that follows.



Approach: one fair pipeline, rigorously validated

Everything held constant except the method — so differences are real

Fair-comparison protocol

One identical training loop for every strategy; only a small 'hook' differs. Same data splits, model, and compute budget.

Reproducibility

Config tiers (smoke/dev/full), global seeding, per-run manifest (git commit + hardware), ≥ 3 seeds for headline numbers.

Independent verification

A torch-free suite of 20 checks proves the maths (aggregation, privacy accountant, robust aggregators) without a GPU.

From simulation to hardware

Same engine runs in a fast simulation and as a real networked system on a laptop + Raspberry Pi.

Does federation work? Yes — the core result

FedAvg sits between training alone and pooling all data — with no data shared

0.32

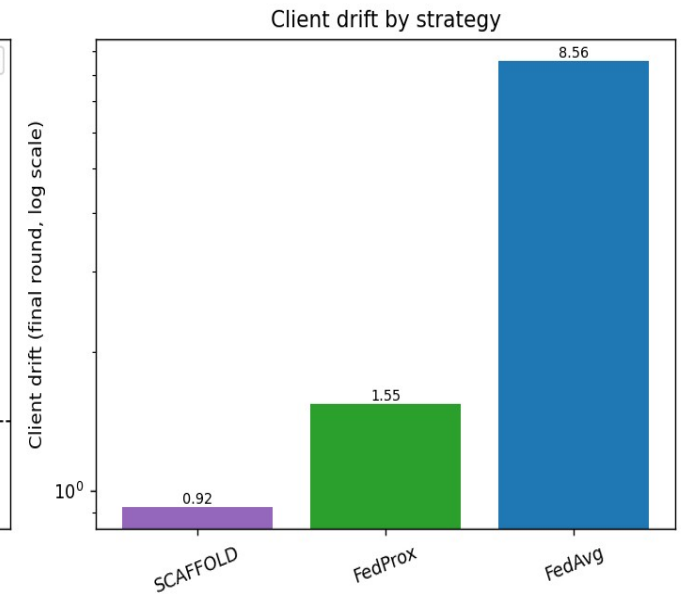
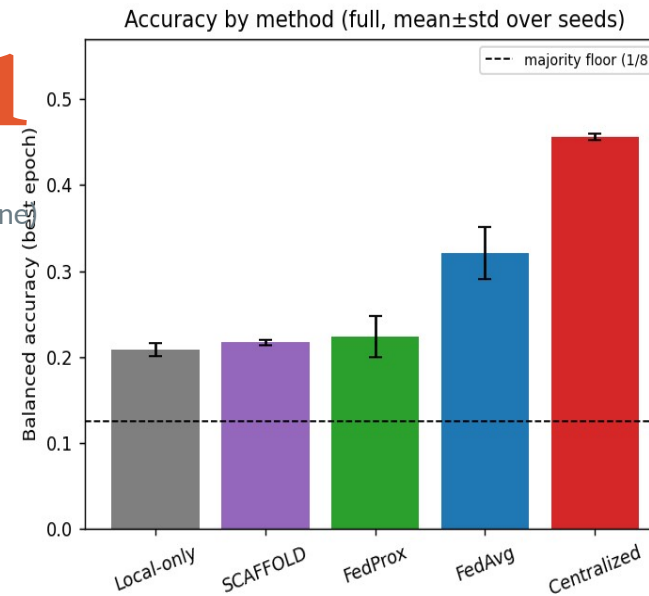
FedAvg (federated)

0.46

Centralized (pooled)

0.21

Local-only (alone)



- FedAvg recovers ~45% of the gap between training alone and pooling everything.
- Bought without any raw image leaving a hospital (answers Q2).
- Client drift is textbook: FedAvg 8.6 $\gg$ FedProx 1.6 $\gg$ SCAFFOLD 0.9.

An honest finding: drift control $\neq$ accuracy

The fancier methods reduce client drift 5–9× exactly as theory predicts — but on this data that did NOT translate into higher accuracy.

FedProx

0.224

proximal term

SCAFFOLD

0.217

control variates

FedAdam

—

adaptive server (unstable)

FedAvg

0.320

the strong baseline

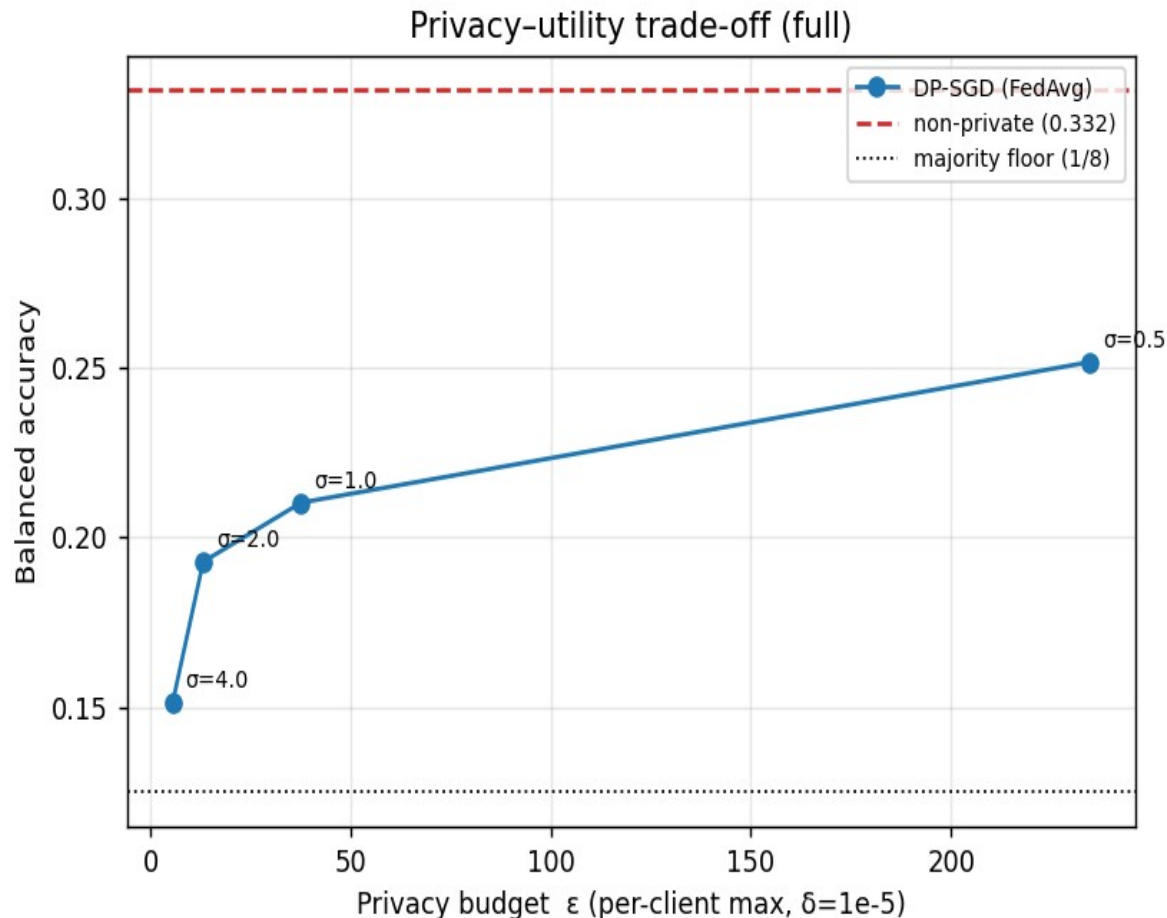
We report this straight: on Fed-ISIC2019, well-tuned FedAvg wins. FedProx/SCAFFOLD deliver stability, not accuracy; FedAdam was too tuning-sensitive to converge. Honesty over hype.

Privacy 1 — a mathematical guarantee (Differential Privacy)

A knob, ϵ , sets the strength: smaller ϵ = more privacy = lower accuracy

- DP adds calibrated noise so no single patient noticeably changes the model.
- We built our OWN privacy accountant (pure Python) to verify ϵ — not trusting the library blindly.
- The curve makes the price explicit: $\epsilon \approx 38$ keeps 0.21; very strong privacy ($\epsilon \approx 5$) drops to 0.15.

Turns 'we added noise' into a quantified promise a decision-maker can dial.



Privacy 2 — proving it works: a membership-inference attack

We didn't just claim privacy — we attacked the model

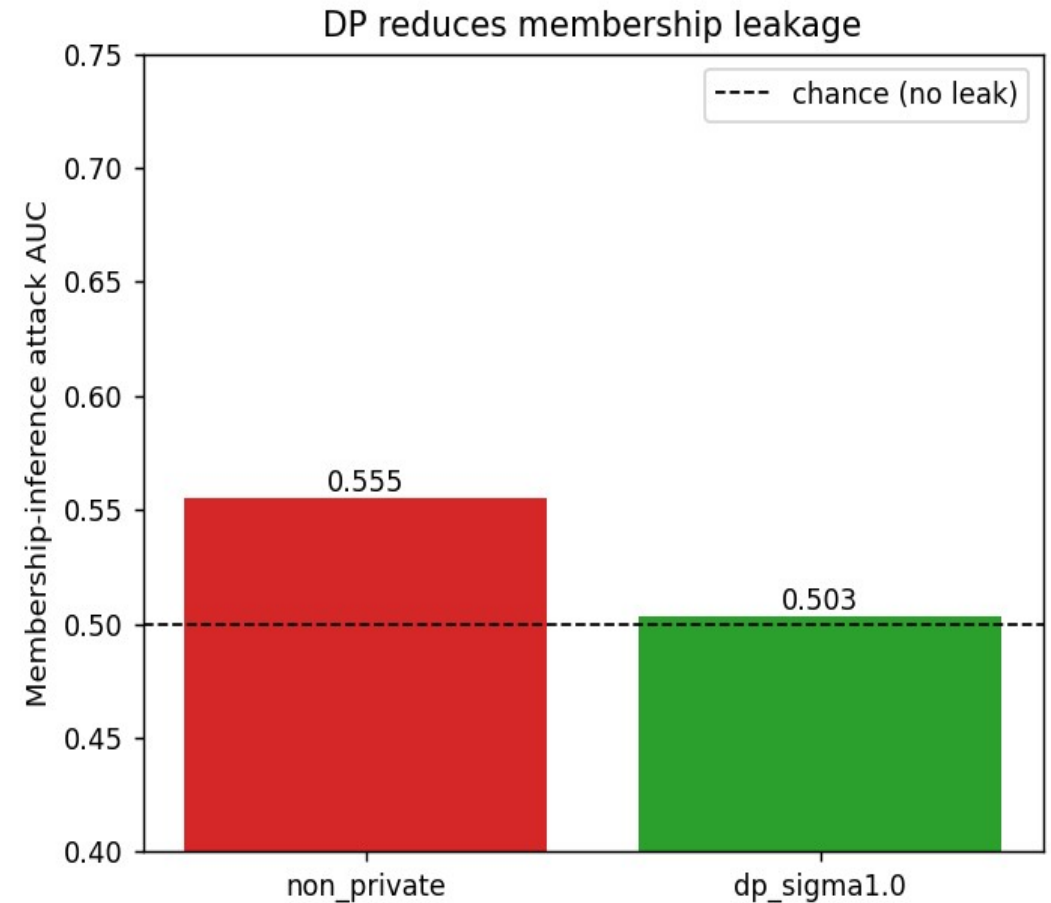
0.55

attack AUC — no privacy (leaks)

0.50

attack AUC — with DP (= chance)

- The attack guesses whether a specific patient was in the training data.
- 0.5 means the attacker learns nothing; above 0.5 means leakage.
- DP measurably shuts the attack down — empirical proof, not a promise.

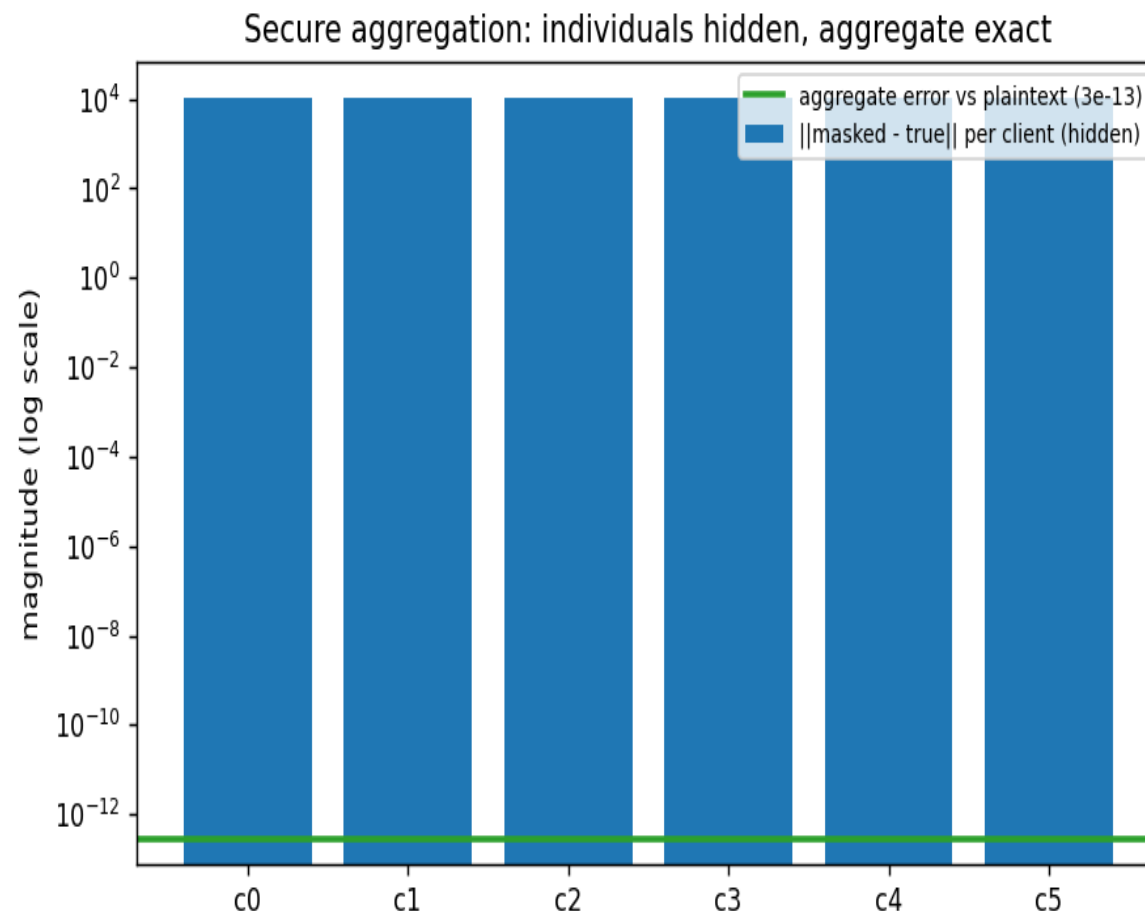


Privacy 3 — hiding each hospital's update (Secure Aggregation)

The coordinator learns the sum, never any single contribution

- Different threat from DP: this hides each hospital's individual update from the server.
- Hospitals add pairwise secret masks that cancel out when summed.
- Verified exact: the recovered average matches plaintext to ~13 decimal places (error $\approx 3 \times 10^{-13}$).

Closes the 'curious server' risk: the coordinator gets the average it needs, but can't inspect any hospital.



Security — what if a hospital is malicious? (Poisoning & defense)

BONUS: the integrity side of security — attack and defend

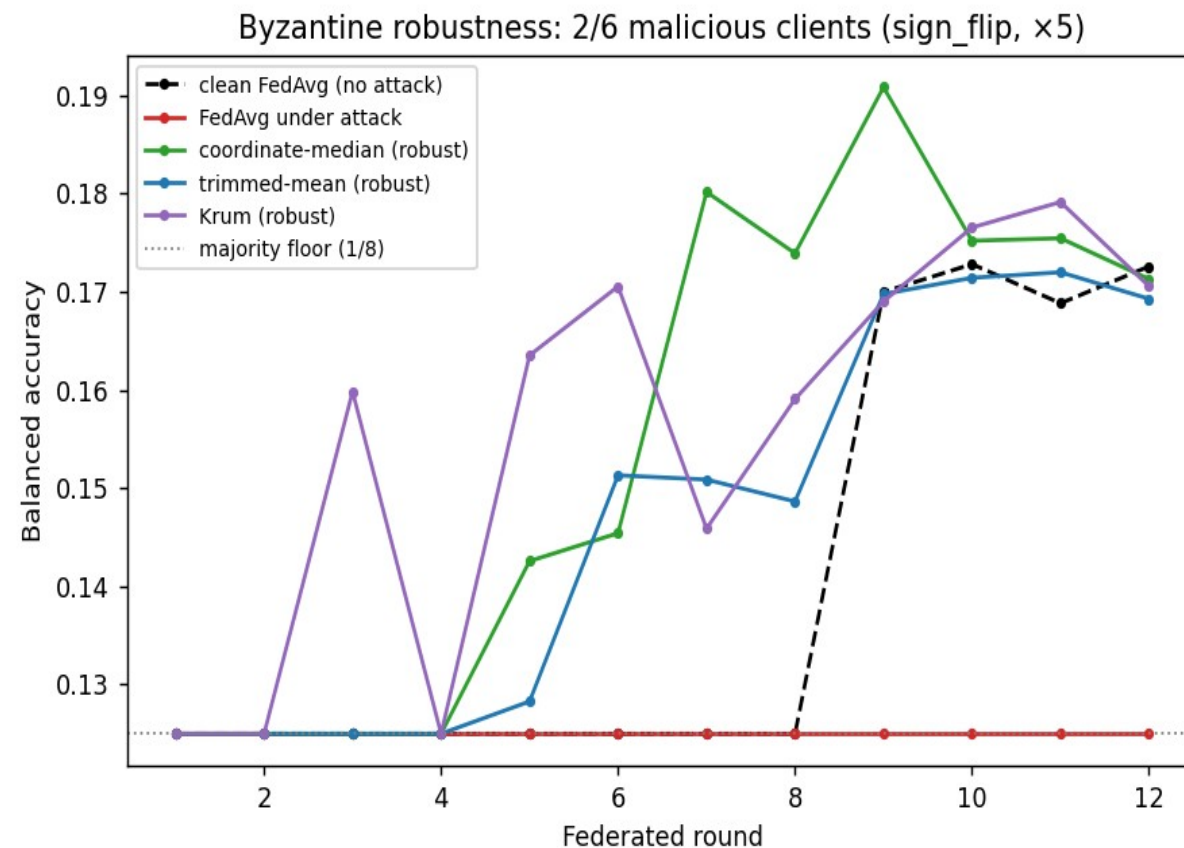
- 2 of 6 hospitals send poisoned updates to sabotage the model.
- Plain FedAvg (a mean) has no defence — it collapses to random (0.125).
- Robust aggregators — median, trimmed-mean, Krum — bound each client's influence and recover (~0.18–0.19).

0.125

FedAvg under attack (random)

0.19

robust aggregation (defended)



Efficiency — cutting network cost 50× (communication)

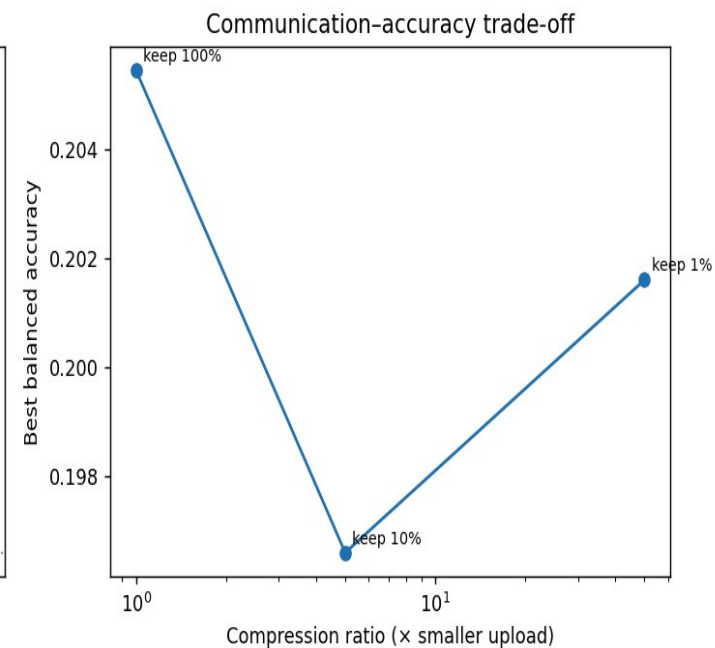
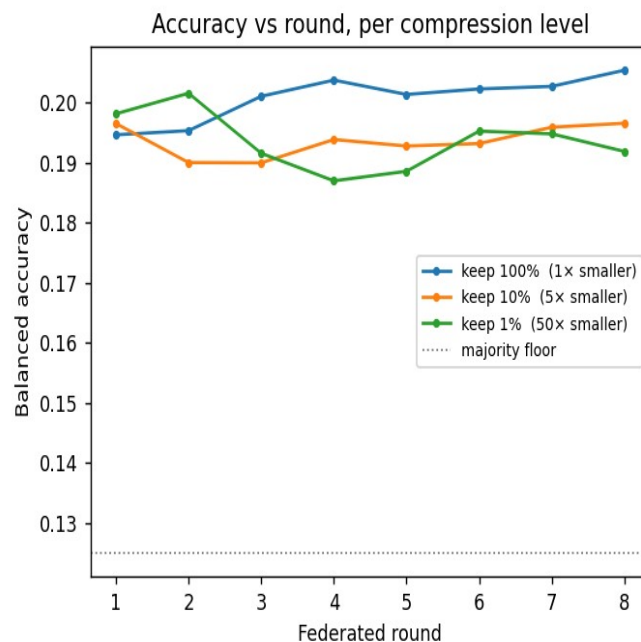
BONUS: FL's real bottleneck is bandwidth, acute for edge devices

50×

smaller uploads

≈ 0

accuracy lost



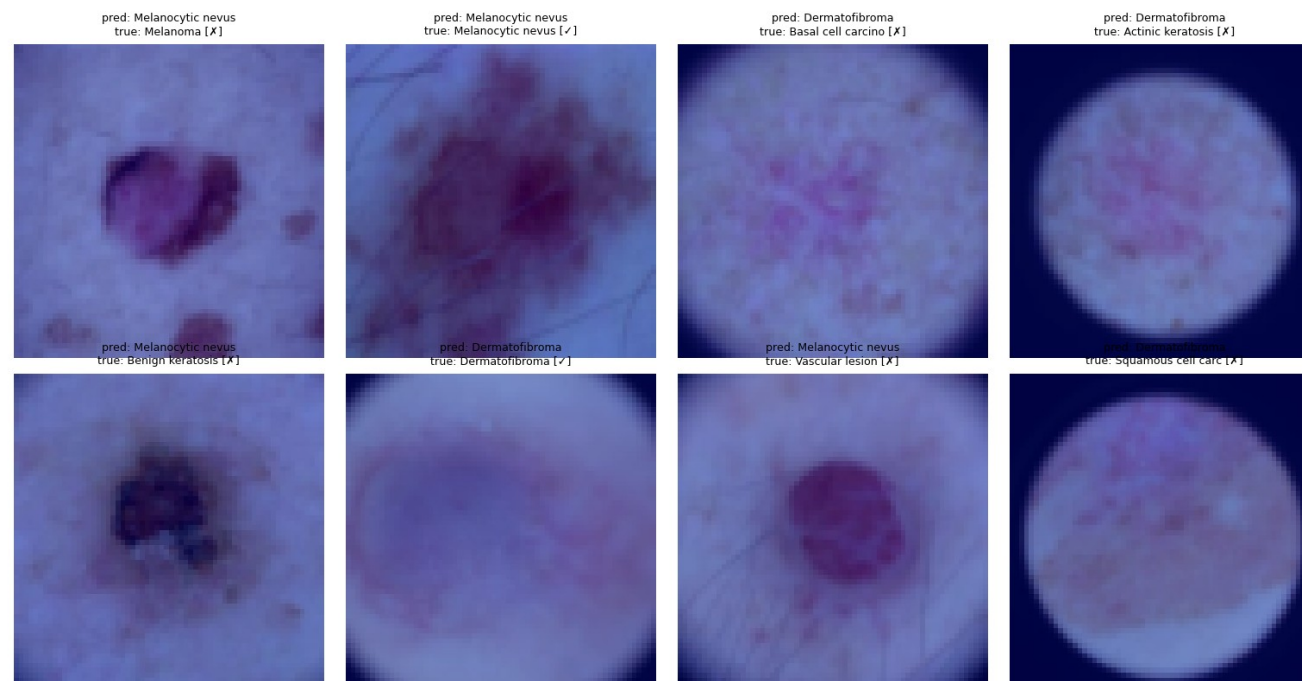
- Send only each hospital's largest 1% of update values (layer-wise top-k), skip the rest.
- 44.7 MB → 0.89 MB per round, with accuracy essentially unchanged (0.202 vs 0.205).
- Pairs with the Raspberry Pi: the slowest, weakest device gains the most from sending less.

Trust — looking inside the model (Explainability)

BONUS: an honest limitation, surfaced deliberately

- Grad-CAM highlights the image regions that drove each decision.
- On our compute-limited model, attention is diffuse and sometimes lands on image borders/artifacts, not the lesion.
- This is a documented 'shortcut' in dermoscopy AI — and catching it is exactly why explainability belongs in the pipeline.

Grad-CAM — where the federated model looks when classifying skin lesions



Reported openly: a production model would need stronger, artifact-free attention before clinical use.

Real hardware: from simulation to a Raspberry Pi

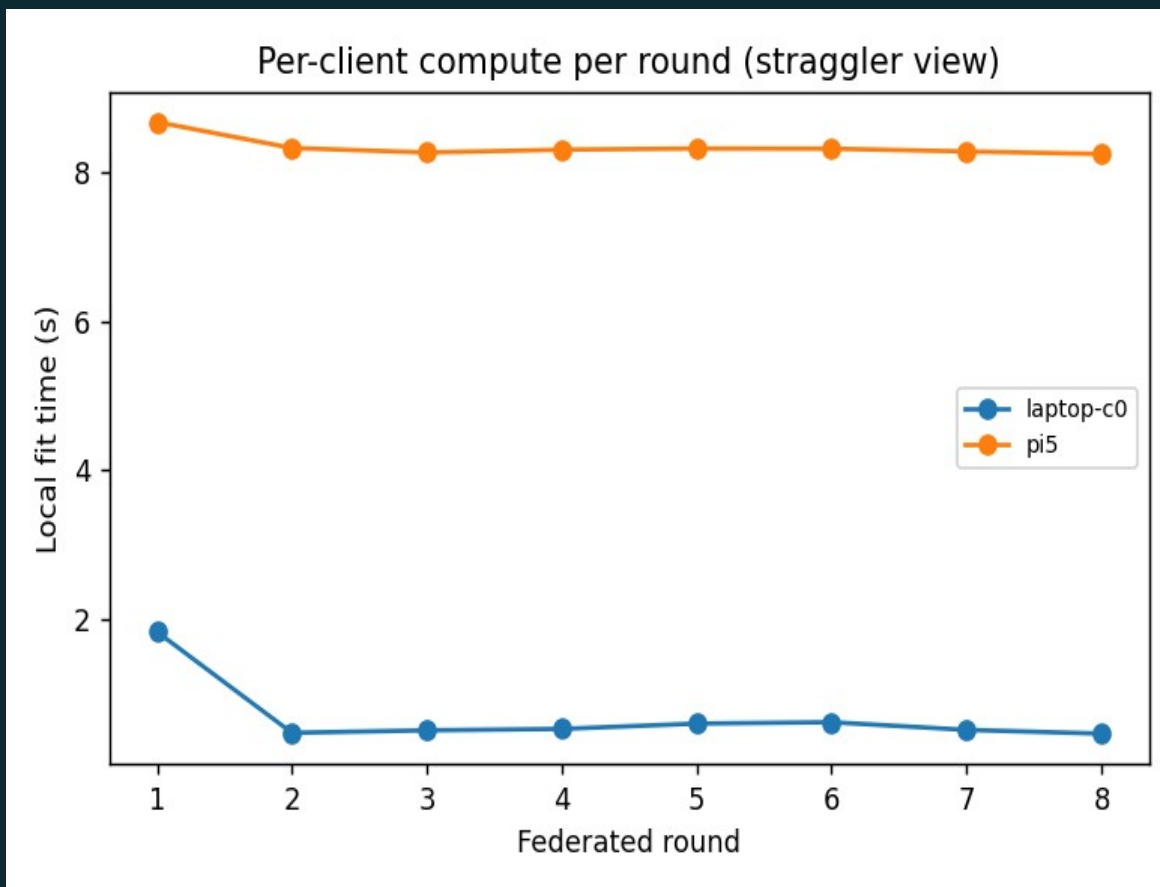
BONUS: genuinely distributed FL — a laptop coordinator + a Raspberry Pi 5 as a real 'edge hospital'

- Only model weights crossed the network — the Pi's images never left it.
- It exposes the straggler problem: the Pi is $\sim 17\times$ slower per round, and synchronous FedAvg waits for it.
- A real systems insight a single-machine simulation cannot show — and why the 50 $\times$ compression matters.

17 $\times$

slower on the Pi

An $\approx$ €80 device meaningfully joins training — with no data leaving it.



Validation, rigour & honesty

Correctness treated as an acceptance requirement, not an afterthought

20 / 20

torch-free maths checks (aggregation, SCAFFOLD, privacy accountant, secure-agg, robust aggregators)

≥ 3 seeds

for every headline number, reported as mean $\pm$ std

Manifests

every run records git commit, tier, and hardware — fully reproducible

**Honest
negatives**

FedAdam and the Grad-CAM limitation reported straight, not hidden

Conclusion — when is FL preferable?

- When data cannot leave an institution, FL beats both training alone and slow, risky data-sharing agreements — it recovered ~45% of the centralized-vs-isolated gap with no raw data leaving a hospital.
- Privacy is layered and verified: differential privacy (with a self-built accountant), a defeated membership attack, and secure aggregation.
- Security covers both sides: confidentiality and integrity (robust aggregation defeats a poisoning attack).
- Practical & real: 50× cheaper communication, demonstrated live on a Raspberry Pi.
- *Future work: EfficientNet backbone for higher accuracy, client-level DP, full-tier robustness runs, artifact-free explainability.*

Thank you

Questions welcome — dashboard, technical report, and code are ready to explore.

`github.com /MahmoudAsadi97/cross-silo-film-medical-image-classification`